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Advanced Digital Design ENCS3310

COURSE PROJECT

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Abstarct

This project presents the design and verification of a finite state machine (FSM) – based traffic light controller implemented using Verilog HDL. The controller manages traffic flow for a main road, a side road, and a pedestrian crossing while strictly satisfying predefined timing and safety constraints. The system responds to vehicle sensor and pedestrian button inputs, enforces a minimum green time for the main road, and provides fixed duration service for the side road and pedestrian crossing. To ensure reliable operation, request signals are latched to prevent missed events, and all state transitions pass through safe yellow and all-red phases. The design is verified using a self-checking testbench that validates timing accuracy, state transitions, and priority handling under multiple scenarios. Simulation results and waveform analysis confirm that the implemented controller operates correctly and meets all project specifications.

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1 System Specification

The traffic light controller is designed to manage an intersection consisting of a main road, a side road, and a pedestrian crossing. The system is modeled as a synchronous finite state machine (FSM) operating on a clock signal and supporting an active-low reset. Its behavior is defined by the following inputs, outputs, states, and timing constraints.

1.1 Inputs

- **clk**: System clock used to synchronize all state transitions.
- **rst**: Active-low reset signal that initializes the controller to a safe default state.
- **VS (Vehicle Sensor)**: Indicates the presence of vehicles waiting on the side road.
- **PB (Pedestrian Button)**: Indicates a pedestrian request to cross the road.

1.2 Outputs

- **light_main [2:0]**: Three-bit signal representing the traffic light of the main road
  - 001 = Green, 010 = Yellow, 100 = Red
- **light_side [2:0]**: Three-bit signal representing the traffic light of the side road
  - 001 = Green, 010 = Yellow, 100 = Red
- **walk**: One-bit pedestrian signal
  - 1 = Walk , 0 = Don't walk

1.3 FSM States

The controller consists of the following 7 states:

- **S_M_GREEN**: Main road green, side road red and walk=0.
- **S_M_YELLOW**: Main road yellow, side road red and walk=0.
- **S_ALL_RED1**: All-red state after main road yellow.
- **S_S_GREEN**: Side road green, main road read and walk=0.
- **S_S_YELLOW**: Side road yellow, main road read and walk=0.
- **S_ALL_RED2**: All-red state before returning to main road.
- **S_WALK**: Pedestrian walk state and side and main are red.

1.4 Timing Constraints

Timing note: My RTL uses a cycle counter. In simulation, durations are verified in **clock cycles** (60, 40, 5, 2 cycles). These cycle counts correspond to the required seconds So all timing is measured in **clock cycles** in my RTL (as if it's the time unit instead of seconds):

- Main road green duration is **at least 60 cycles**.
- Side road green duration is **exactly 40 cycles**.
- Pedestrian walk duration is **exactly 40 cycles**.
- Yellow light duration is **5 cycles**.
- All-red state duration is **2 cycles**.

1.5 Functional Behavior

1. After reset, the system starts in **S_M_GREEN**, giving priority to the main road.
2. If **VS** or **PB** is asserted, the controller completes the minimum main green time before transitioning.
3. The transition sequence for a request is:
   Main Green → Main Yellow → All Red → Side Green or Walk.
4. Pedestrian requests are given priority over side road vehicles.
5. Requests occurring during side-road green are latched and served in the next cycle.
6. Safe operation is guaranteed by enforcing yellow and all-red states between conflicting

2 State Encoding

The traffic light controller uses **binary state encoding**. Each FSM state is represented using a 3-bit binary value. This encoding choice is suitable because the number of states is small, and binary encoding minimizes hardware resources compared to one-hot encoding.

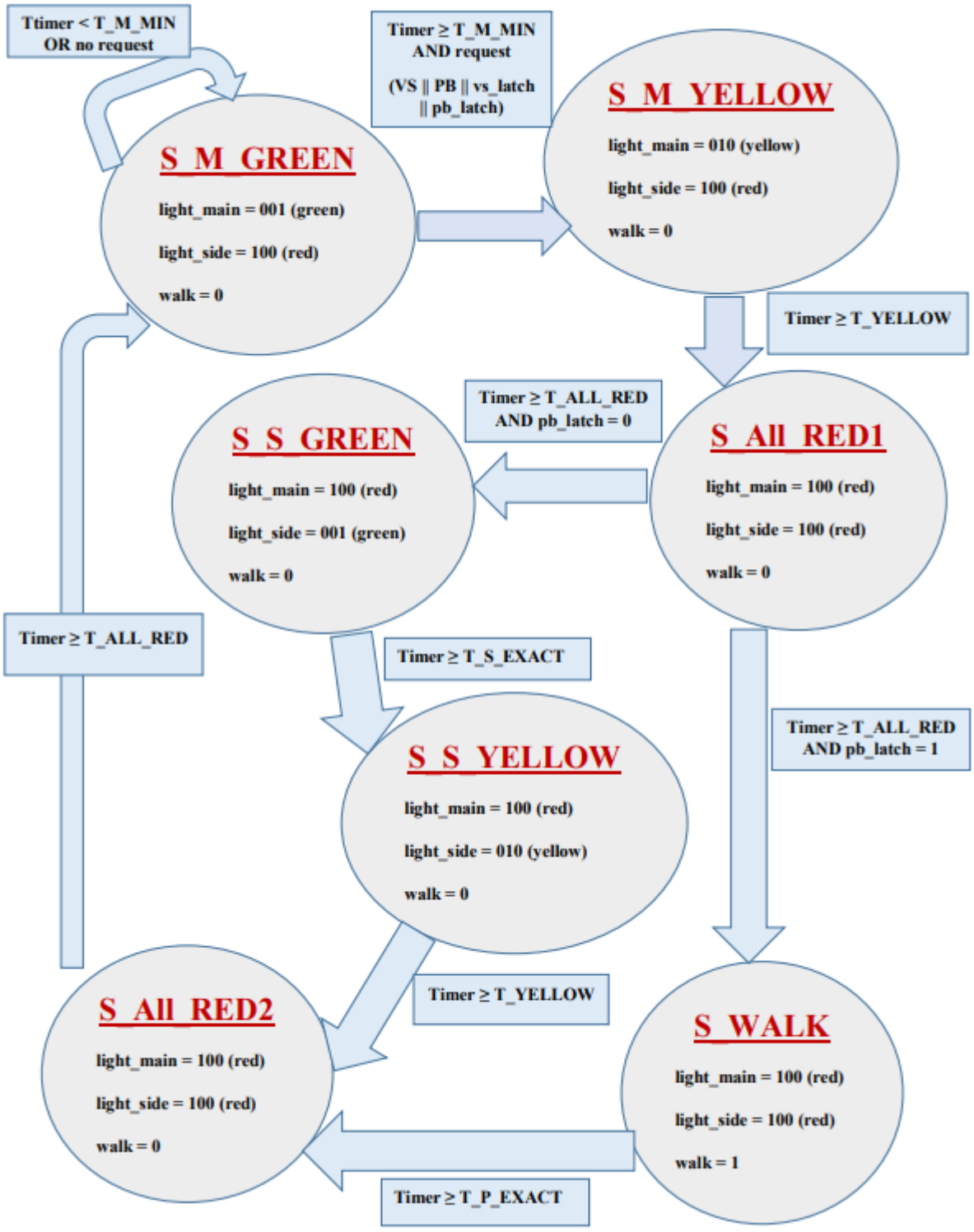
The state encoding is defined as follows:

- **S_M_GREEN = 3'd0**
- **S_M_YELLOW = 3'd1**
- **S_ALL_RED1 = 3'd2**
- **S_S_GREEN = 3'd3**
- **S_S_YELLOW = 3'd4**
- **S_ALL_RED2 = 3'd5**
- **S_WALK = 3'd6**

The FSM uses two registers:

- **curr_state**: holds the current active state
- **next_state**: holds the next state computed by the transition logic

3 FSM State Diagram Description



The figure shows the finite state machine (FSM) of the traffic light controller. The controller starts in the **Main Road Green state (000)**, which is the default state after reset. The main road remains green for a minimum time before responding to any request. When a vehicle sensor (VS) or pedestrian button (PB) is asserted, the FSM completes the main green phase and transitions through **Main Yellow (001)** followed by an **All-Red state (010)** to ensure safe switching.

From the all-red state, the FSM either enters the **Side Road Green state (011)** if the request is from the vehicle sensor, or the **Pedestrian Walk state (110)** if a pedestrian request exists. Both the side green and walk states last for fixed, exact durations. The FSM then transitions through **Side Yellow (100)** and a second **All-Red state (101)** before returning to the **Main Road Green state**, completing one full traffic cycle.

Pedestrian requests that occur during the Side Road Green state are not served immediately. Instead, they are latched and queued to be handled in the next cycle after the FSM returns to the main road phase, as required by the specification.

All traffic light outputs are determined solely by the current state (Moore machine), while input signals (VS and PB) affect only the state transitions. This design ensures deterministic timing, safe transitions, and correct prioritization of pedestrian requests.

4 Timing Behavior

The timing behavior of the traffic light controller is implemented using a synchronous counter that measures the number of clock cycles spent in each FSM state. A dedicated timer register is incremented on every rising edge of the clock as long as the controller remains in the same state. When a state transition occurs, the timer is reset to zero, ensuring accurate measurement for each state duration.

Each state compares the timer value against its corresponding timing parameter to determine when a transition should occur. For states with **exact durations** (side road green, pedestrian walk, yellow, and all-red), the FSM transitions when the timer reaches the specified value. For the **main road green state**, the controller enforces a minimum duration; after this minimum time elapses, the FSM transitions only if a valid request from the vehicle sensor or pedestrian button exists.

Because the timer starts counting from zero, the comparison is performed using $(T - 1)$ to guarantee that each state lasts for the exact number of required clock cycles. This approach ensures precise timing control and avoids off-by-one errors.

All timing is therefore deterministic, fully synchronous, and independent of input signal glitches, since requests are latched and evaluated only through the FSM transition logic.

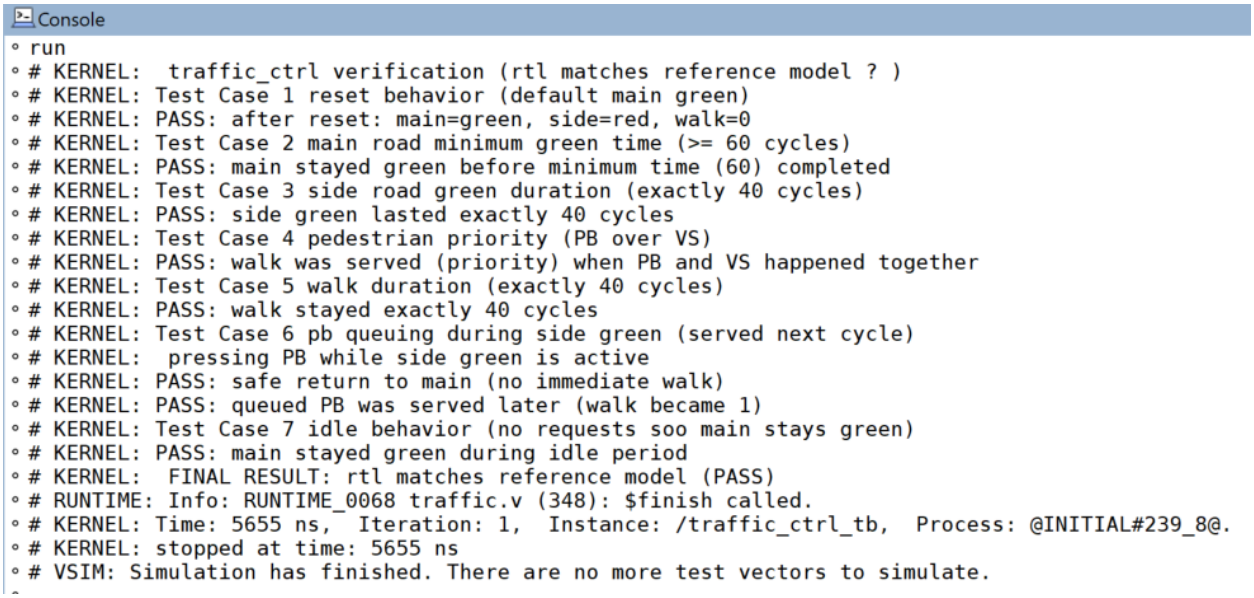
5 Clock Assumptions and Operation

The traffic light controller is a fully synchronous finite state machine driven by a single clock signal. All state transitions, timer updates, and request latching occur on the rising edge of the clock, while the active-low reset initializes the system to a safe default state.

In my project, **time is measured in clock cycles**, and each clock cycle is treated as one unit of time. All timing requirements specified in the project—such as minimum green time, yellow duration, all-red duration, and pedestrian walk time—are implemented using cycle counts rather than real seconds.

This approach follows the project specification, which defines timing behavior purely in terms of clock cycles. By using cycle-based timing, the FSM behavior becomes deterministic and easy to verify in simulation, while remaining independent of any specific real-world clock frequency.

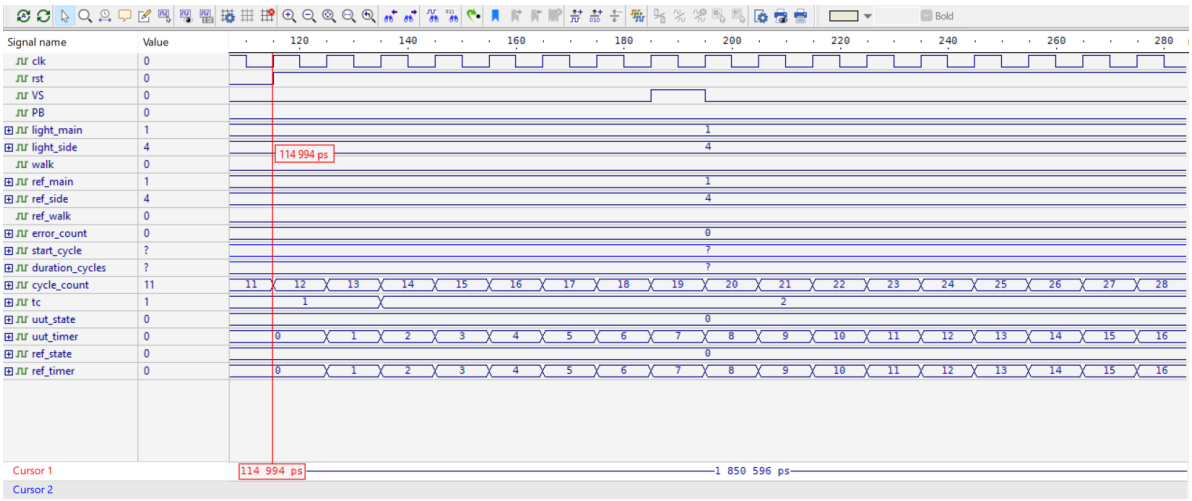
6 Verification Results and Console Output

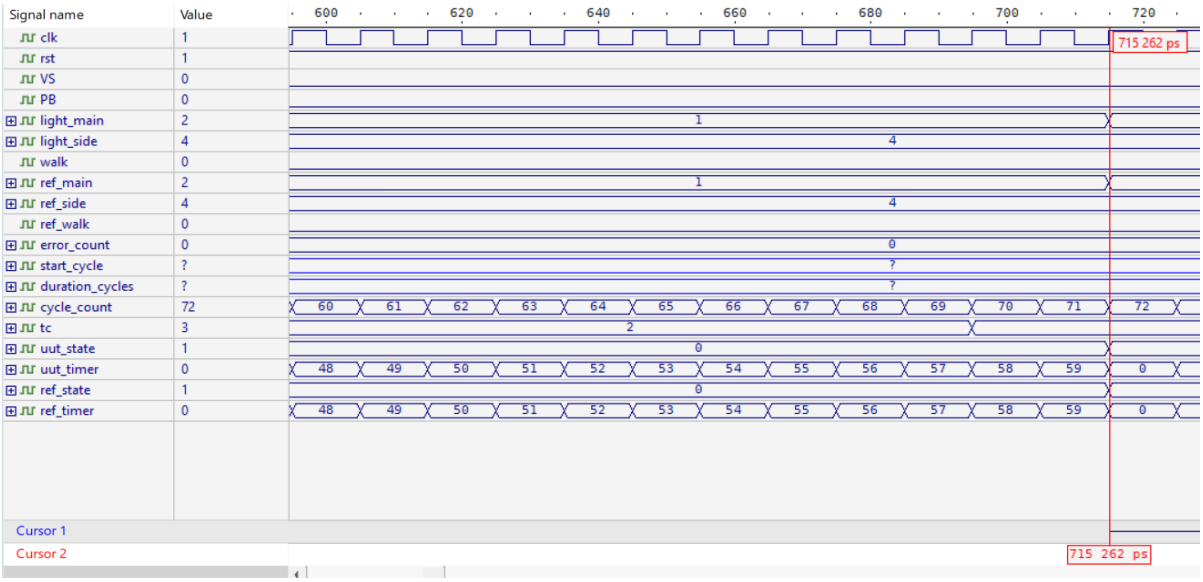


As shown in the console figure, the design was verified using a self checking testbench that continuously compares the RTL outputs with a reference model during simulation. The console output shows that all seven test cases were executed successfully, covering reset behavior, timing constraints, request prioritization, and idle operation. Each test case prints a clear message indicating whether the expected behavior was achieved. The results confirm that the main road remains green for the required minimum time, the side road and pedestrian walk states last exactly the specified number of clock cycles, and pedestrian requests are correctly prioritized and safely queued when needed. Since no mismatches were detected throughout the simulation, the final message confirms that the RTL implementation fully matches the reference model and behaves exactly as intended.

7 Simulation and Testing

7.1 Test Case 1: Main Road Minimum Green Duration





After the reset signal is deasserted, the traffic light controller enters its default state **S_M_GREEN**, in which the main road light is green while the side road and pedestrian signals remain red. As shown in the waveform, this behavior starts at **cycle 12**, which represents the first stable clock cycle following reset.

From **cycle 12 to cycle 72**, the controller remains continuously in the **Main Road Green** state. During this interval, the signal light_main stays green, light_side remains red, and the pedestrian signal walk stays low. Internally, the FSM state (uut_state = 0) does not change, and the timer

(uut_timer) increments on each rising edge of the clock from **0 to 59**, resulting in a total duration of **60 clock cycles**.

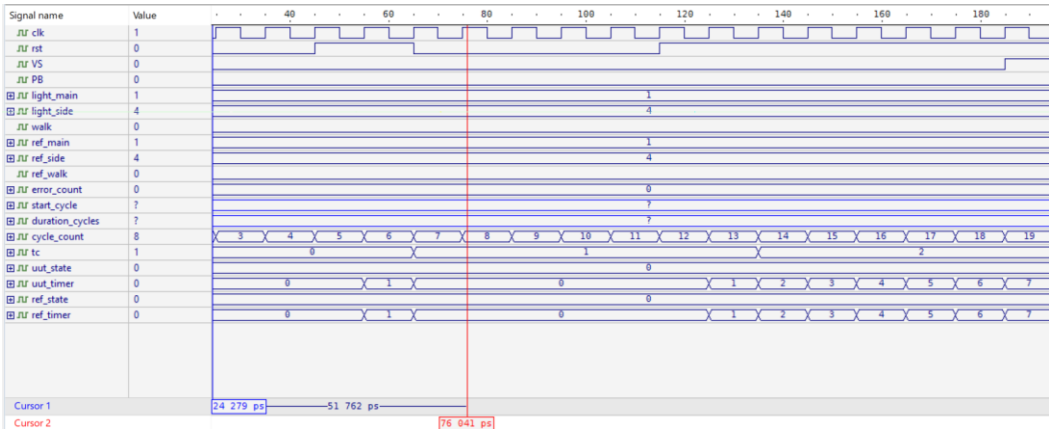
At **cycle 72**, the timer reaches its terminal value and resets, triggering a transition to **the** Main Yellow state. This confirms that the controller correctly enforces the minimum green duration requirement before responding to any pending requests.

The waveform also shows a perfect match between the RTL implementation and the reference model throughout this test case, with no mismatches detected, verifying correct timing behavior and functional correctness.

****NOTE:**

The signals start_cycle and duration_cycles are testbench variables used only for measuring timing during specific test cases. They appear as unknown (?) in the waveform before being assigned, because they are updated only when the relevant event occurs (for example, when a green or walk phase starts or ends). This behavior is expected and confirms that timing measurements are performed dynamically based on actual FSM transitions rather than being pre-initialized.

7.2 Test Case 2: Reset Behavior

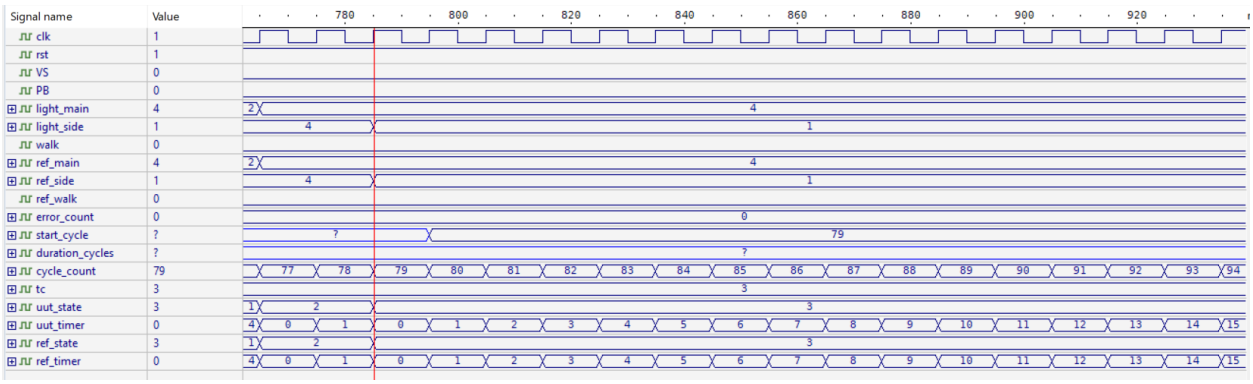


When the active-low reset signal (rst = 0) is asserted, the traffic light controller is forced into a safe initial state. The FSM enters the S_M_GREEN state, enabling the main road green light while keeping the side road red and the pedestrian walk signal red. All internal timers and request latches are cleared.

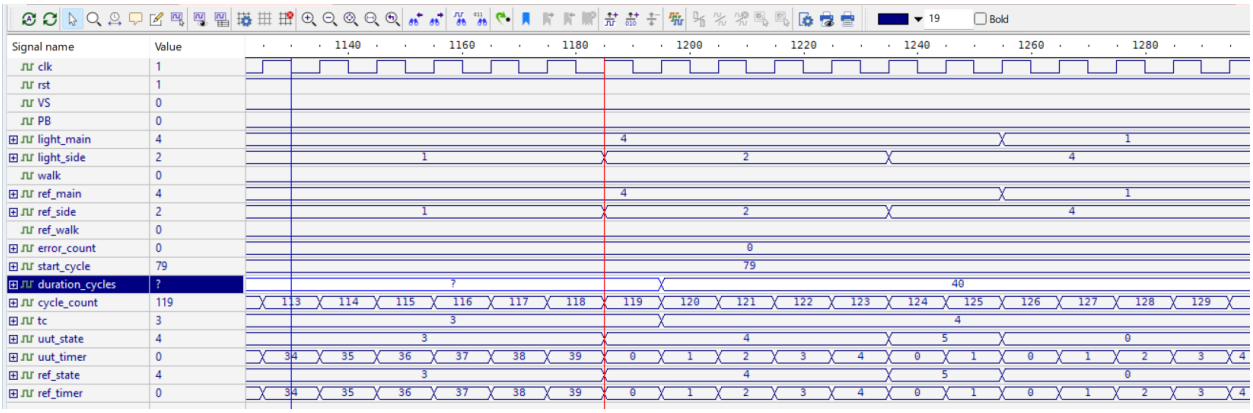
After reset is de-asserted (rst = 1), the FSM resumes normal operation synchronously with the clock. The controller remains in the main green state and the internal timer begins counting clock cycles. Since no vehicle or pedestrian requests are present, no state transition occurs immediately. The identical behavior of both the RTL design and the reference model confirms correct reset handling and initialization.

7.3 Test Case 3 : Side road green duration (exactly 40 cycles)

Start of side green :



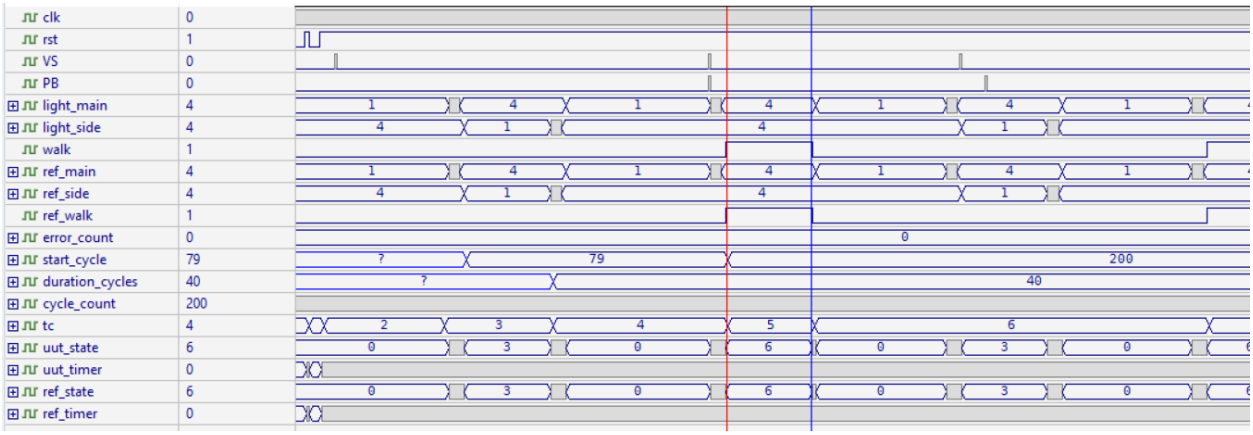
End of side green:



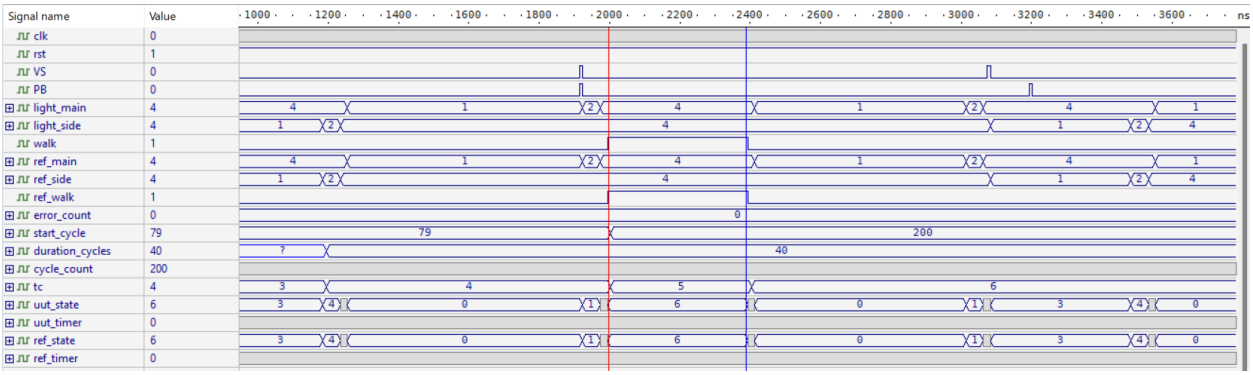
The side road green interval starts at **cycle 79** and ends at **cycle 119**, resulting in a total duration of **40 clock cycles**, which exactly matches the specified timing requirement. During this period, the internal timer (uut_timer) increments from 0 to 39, confirming that the FSM remains in the side green state for the correct number of cycles. After the 40-cycle interval completes, the FSM transitions to the Side Yellow state, followed by the all-red state before returning to the main road.

The reference model signals (ref_main, ref_side, and ref_state) match the RTL outputs throughout this interval, and no mismatches are detected. This confirms that the side road green timing is implemented correctly and behaves as required by the specification.

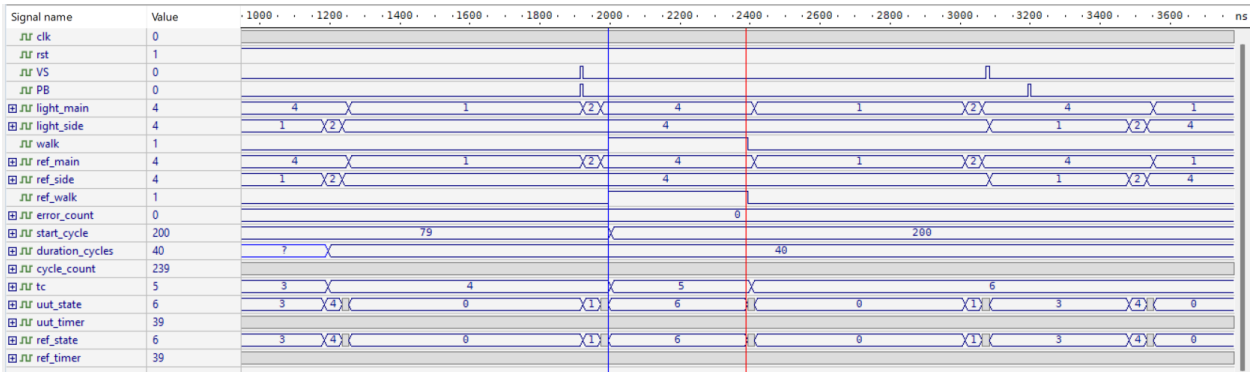
7.4 Test Case 4: Pedestrian priority + walk duration



Start of walk green (1):



End of walk green:



This test case verifies that the traffic light controller correctly prioritizes pedestrian requests over vehicle requests and enforces the exact walk duration specified in the design.

As shown in the waveform, both the vehicle sensor (VS) and the pedestrian button (PB) are asserted while the system is in the Main Road Green state. The FSM does not react immediately; instead, it first completes the minimum required green time for the main road. After this minimum time elapses, the controller transitions sequentially through Main Yellow and All-Red, ensuring a safe state change.

At the all-red decision point, the FSM correctly selects the Pedestrian Walk state instead of the side road green state. This behavior confirms that **pedestrian requests have higher priority than vehicle requests**, as required by the specification. During the walk state, both road signals remain red while the walk output is asserted.

Using the cycle counter and waveform cursors, the walk signal is observed to remain active for **exactly 40 clock cycles**. After the walk duration completes, the controller transitions through the second all-red state and then safely returns to the Main Road Green state.

Throughout this entire sequence, the RTL outputs (light_main, light_side, walk) match the reference model outputs exactly, and no mismatches are detected. This confirms both **correct** prioritization logic and accurate timing behavior for pedestrian handling.

Conclusion

This project presented the design and verification of a traffic light controller using a synchronous finite state machine. The FSM was carefully derived from the given specification and implemented using binary state encoding and cycle-based timing control.

Simulation and self-checking testbench results confirm that the controller satisfies all functional and timing requirements, including safe transitions, correct prioritization of pedestrian requests, and deterministic behavior. The design is modular, readable, and easily extensible, making it suitable for further enhancements or hardware implementation.